COMPUTER NETWORKING

Comprehensive Study Notes

Fundamentals • Network Types & Ranges • How Networks Operate • LAN & WAN • Network Topologies

1. What Is Networking?

Computer networking is the practice of connecting two or more computing devices (computers, servers, phones, printers, sensors, etc.) so that they can share resources, exchange data, and communicate with one another. A network can be as small as two laptops linked by a cable, or as large as the global Internet connecting billions of devices.

The purpose of networking is to allow devices to **share information and resources** such as files, printers, internet access, and applications, and to enable **communication** (email, messaging, video calls) and **centralized management** of data and security.

Core Building Blocks of a Network

- **Node / Host:** Any device connected to the network (computer, printer, phone, server, IoT device).
- **Link (Medium):** The physical or wireless path data travels on – copper cable, fiber-optic cable, or radio waves (Wi-Fi).
- **Protocol:** A set of rules that governs how data is formatted, transmitted, and received (e.g. TCP/IP, HTTP, FTP).
- **Network Interface Card (NIC):** Hardware that lets a device physically connect to a network.
- **Router:** Directs data packets between different networks, choosing the best path.
- **Switch:** Connects devices within the same network and forwards data only to the intended recipient.
- **Hub:** A simple, older device that broadcasts incoming data to all connected devices (largely obsolete).
- **Modem:** Converts digital signals from your devices into signals that can travel over telephone, cable, or fiber lines, and back again.
- **IP Address:** A unique logical address (e.g. 192.168.1.10) that identifies a device on a network.
- **MAC Address:** A unique physical address burned into a NIC that identifies a device at the hardware level.

Why We Classify Networks

Networks are usually classified in two major ways: by their **geographic size/range** (PAN, LAN, MAN, WAN, etc.) and by their **physical or logical arrangement**, called a **topology** (bus, star, ring, mesh, tree, hybrid). Understanding both dimensions is essential to designing, troubleshooting, and securing any network.

2. How Networks Operate

At its core, a network moves data from a **source** device to a **destination** device. This happens in several coordinated steps, governed by layered protocol models such as the **OSI Model** (7 layers) and the more commonly used **TCP/IP Model** (4 layers).

2.1 The Journey of Data: Packet Switching

Most modern networks (including the Internet) use **packet switching**. Data is broken into small units called **packets**, each containing a piece of the original data plus header information (source address, destination address, sequence number). Packets can travel independently across different paths and are reassembled in the correct order at the destination. This makes networks efficient, resilient to failures, and able to share bandwidth among many users at once.

2.2 Addressing and Routing

- **IP Addressing:** Every device gets a logical IP address (IPv4 like 192.168.1.5, or IPv6) that identifies it on a network and allows routers to know where to send data.
- **MAC Addressing:** Within a local network, devices are also identified by physical MAC addresses used by switches for delivery.
- **Routing:** Routers examine the destination IP address of each packet and forward it along the best available path toward its destination, possibly through many intermediate routers (hops).
- **Switching:** Within a LAN, switches use MAC address tables to forward frames directly to the correct device instead of broadcasting to everyone.
- **DNS (Domain Name System):** Translates human-friendly names (e.g. www.example.com) into IP addresses that devices can route to.

2.3 Protocols That Make It Work

Protocol	Layer / Purpose
TCP (Transmission Control Protocol)	Reliable, ordered, connection-based delivery of data (used by web, email, file transfer)
UDP (User Datagram Protocol)	Fast, connectionless delivery with no guarantee of order (used by video streaming, gaming, V
IP (Internet Protocol)	Addressing and routing of packets across networks
HTTP / HTTPS	Transfers web pages between browsers and servers (HTTPS is encrypted)
FTP	Transfers files between computers
DHCP	Automatically assigns IP addresses to devices on a network
DNS	Resolves domain names into IP addresses

2.4 Transmission Modes

- **Simplex:** Data flows in one direction only (e.g. a keyboard sending input, a TV broadcast).

- **Half-Duplex:** Data flows both ways, but only one direction at a time (e.g. a walkie-talkie).
- **Full-Duplex:** Data flows both ways simultaneously (e.g. a telephone call, modern Ethernet).

In short: a network operates by taking data, breaking it into packets, addressing those packets, and using switches and routers — guided by protocols — to move each packet across physical or wireless links until it reaches its destination, where it is reassembled into the original data.

3. Types of Networks and Their Ranges

Networks are commonly classified by their geographic coverage. The diagram below shows how each network type nests within a larger one, from a few centimeters (PAN) up to global scale (WAN).

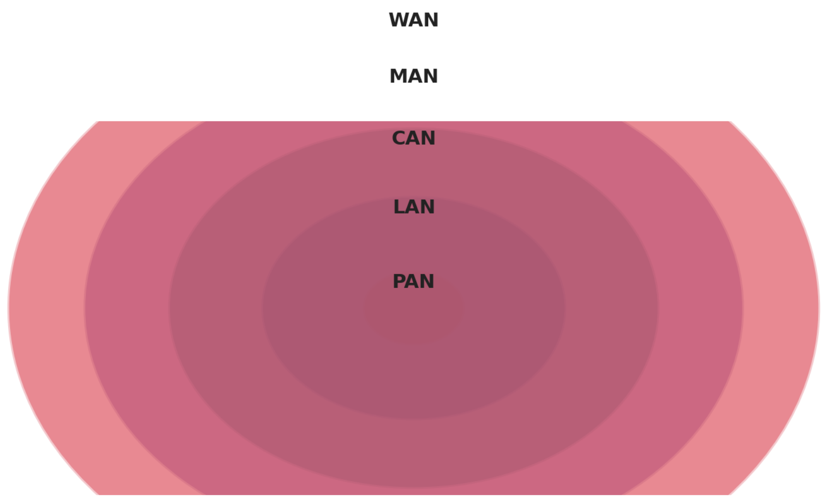

Figure 1: Network types by increasing geographic range

Type	Full Name	Typical Range	Example
PAN	Personal Area Network	Up to ~10 m	Bluetooth earbuds, smartwatch to phone, USB devices
LAN	Local Area Network	10 m – 1 km	Home Wi-Fi, office network, school computer lab
WLAN	Wireless Local Area Network	Same as LAN, wireless	Office/home Wi-Fi network
CAN	Campus Area Network	1 – 5 km	University campus, corporate business park
MAN	Metropolitan Area Network	5 – 50 km	City government network, cable TV network
WAN	Wide Area Network	50+ km, national/global	The Internet, a multinational company's network
SAN	Storage Area Network	Data-center scale	High-speed network linking servers to storage devices

3.1 Local Area Network (LAN)

A **LAN** connects devices within a small, confined area – a single building, home, or office. LANs are typically owned, controlled, and managed by a single organization or individual.

How a LAN Operates

- Devices connect via Ethernet cables or Wi-Fi to a central **switch** or **wireless access point**.
- The switch learns the MAC address of each connected device and forwards traffic only to the intended recipient, minimizing collisions.
- A **router** (often combined with a modem in home setups) connects the LAN to external networks such as the Internet, and typically runs DHCP to assign private IP addresses (e.g. 192.168.x.x) to each device.

- Because distances are short, LANs achieve very high speeds (100 Mbps to 10+ Gbps) and very low latency.

Key Characteristics

- High speed, low latency, low cost per connection.
- Single organization ownership – easy to secure and manage centrally.
- Common technologies: Ethernet (IEEE 802.3), Wi-Fi (IEEE 802.11).

3.2 Wide Area Network (WAN)

A **WAN** spans a large geographic area – connecting multiple LANs across cities, countries, or continents. The Internet is the largest and most well-known WAN in existence.

How a WAN Operates

- WANs link geographically distant LANs using long-distance transmission media such as fiber-optic cables, leased telecom lines, microwave links, or satellite links.
- Traffic is carried through many intermediate **routers** operated by different Internet Service Providers (ISPs) and backbone carriers, each forwarding packets closer to their destination (multi-hop routing).
- Technologies like **MPLS** (Multiprotocol Label Switching), **leased lines**, and **VPNs** (Virtual Private Networks) are used by organizations to securely connect branch offices over a WAN.
- Because signals travel much farther and through shared, third-party infrastructure, WANs generally have **lower speed and higher latency** than LANs, and are more expensive to build and maintain.

Key Characteristics

- Covers large distances – national or global scale.
- Usually owned and operated by multiple organizations/ISPs, not a single entity.
- Relatively slower and higher-latency compared to a LAN, though modern fiber backbones are very fast.
- Examples: the Internet, a bank's network linking branches nationwide, MPLS corporate networks.

3.3 Other Notable Types

- **MAN (Metropolitan Area Network):** Bridges the gap between LAN and WAN, typically covering a city; often used by municipalities or cable/ISP companies to link multiple LANs across town.
- **PAN (Personal Area Network):** The smallest network scale, connecting personal devices within a few meters – e.g. Bluetooth headphones, fitness trackers.
- **CAN (Campus Area Network):** Connects multiple LANs within a limited geographic area like a university or corporate campus, usually via a private backbone.

4. Network Topologies (Configurations)

A **topology** describes how devices (nodes) and cables (links) are physically or logically arranged within a network. The choice of topology affects cost, reliability, scalability, and ease of troubleshooting.

4.1 Bus Topology

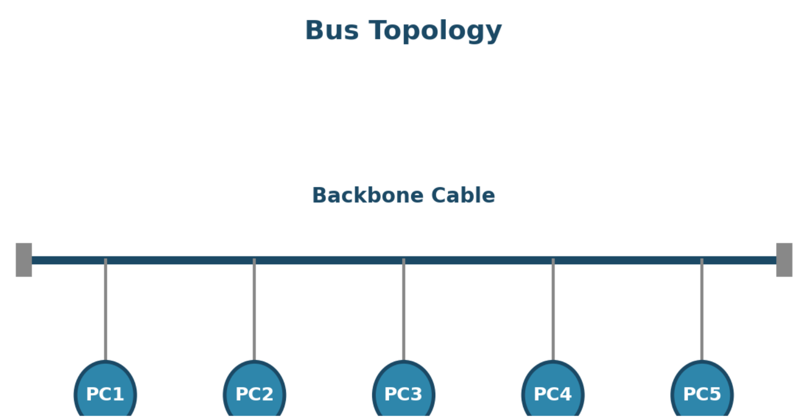

Figure 2: All devices share a single backbone cable

All devices are connected to a single central cable, called the **backbone** or **bus**. Data sent by one device travels along the cable in both directions until it reaches its destination; **terminators** at each end of the cable absorb the signal to prevent it from bouncing back.

- **Advantages:** Simple, inexpensive, uses the least amount of cable.
- **Disadvantages:** A break anywhere in the backbone brings down the entire network; performance degrades as more devices are added; difficult to troubleshoot.
- **Use case:** Mostly historical – early Ethernet (10BASE2/10BASE5) networks. Rarely used today.

4.2 Star Topology

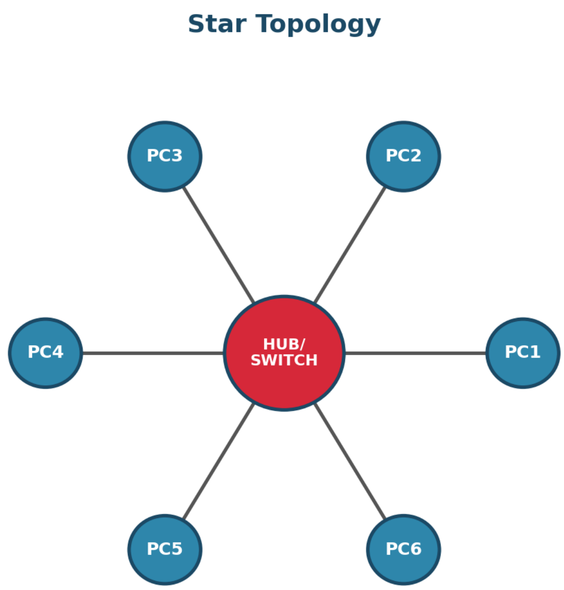

Figure 3: Every device connects to a central hub/switch

Every device connects individually to a central device – typically a **switch** or **hub**. All data passes through this central point, which forwards it to the correct destination device.

- **Advantages:** Easy to install and manage; a single cable failure only affects one device; easy to add or remove devices.
- **Disadvantages:** If the central switch/hub fails, the entire network goes down; requires more cabling than a bus.
- **Use case:** The most common topology today – used in almost all modern homes, offices, and school networks.

4.3 Ring Topology

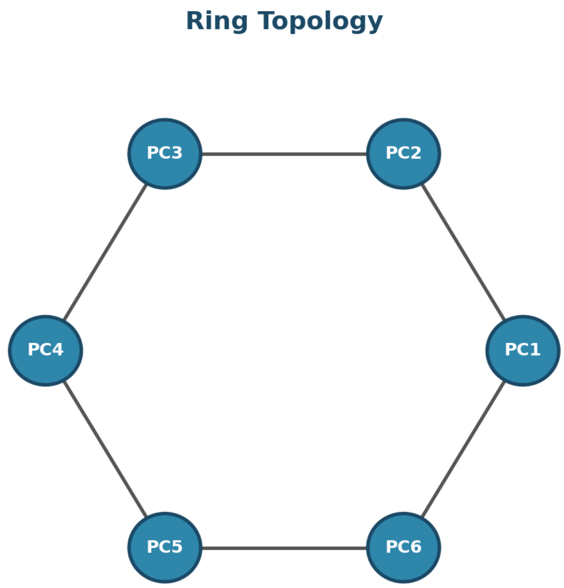

Figure 4: Each device connects to exactly two neighbors, forming a closed loop

Each device connects to exactly two other devices, forming a continuous loop. Data travels around the ring, usually in one direction, passing through each device until it reaches its destination (a technique often called **token passing** is used to control access to the ring).

- **Advantages:** Data flows in an orderly, predictable manner; performs well under heavy load compared to bus.
- **Disadvantages:** A single cable or device failure can disrupt the entire ring (unless a dual/redundant ring is used); adding/removing devices disrupts the network.
- **Use case:** Token Ring networks (legacy), some fiber-optic metropolitan and telecom backbone rings (e.g. SONET).

4.4 Mesh Topology

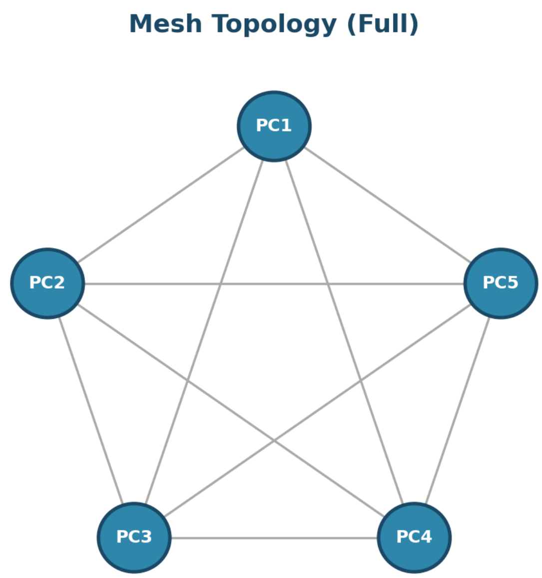

Figure 5: In a full mesh, every device connects directly to every other device

Every device connects directly to every other device. In a **full mesh**, all nodes are interconnected; in a **partial mesh**, only some critical nodes have multiple redundant connections.

- **Advantages:** Extremely reliable and fault-tolerant – if one link fails, data can take another path; no single point of failure.
- **Disadvantages:** Very expensive and complex to install/maintain due to the large amount of cabling required (grows rapidly as devices are added); difficult to manage.
- **Use case:** Critical backbone networks, military and financial networks, wireless mesh networks (mesh Wi-Fi systems), the core of the Internet itself.

4.5 Tree (Hierarchical) Topology

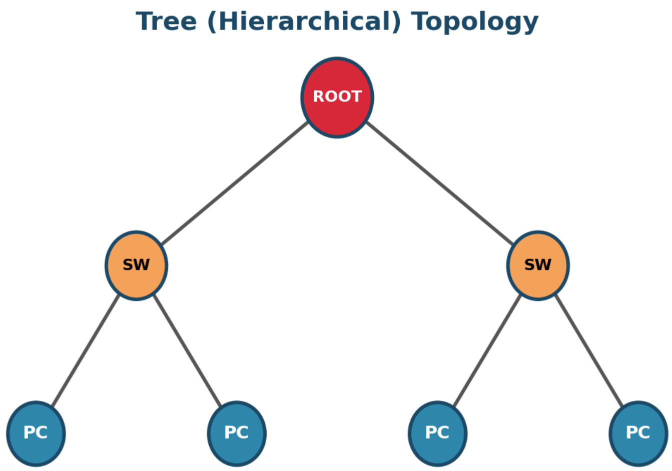

Figure 6: A hierarchy of star networks linked through a root node

A hybrid of bus and star topologies, arranged as a hierarchy: a central **root** node connects to secondary hub/switch nodes, which in turn connect to individual devices – much like the branches of a tree.

- **Advantages:** Scalable and easy to expand by adding new branches; faults in one branch don't necessarily affect others; supported by most networking hardware and software vendors.
- **Disadvantages:** Heavily dependent on the root/backbone node – if it fails, entire branches lose connectivity; can require significant cabling.
- **Use case:** Large corporate and campus networks organized by department or building.

4.6 Hybrid Topology

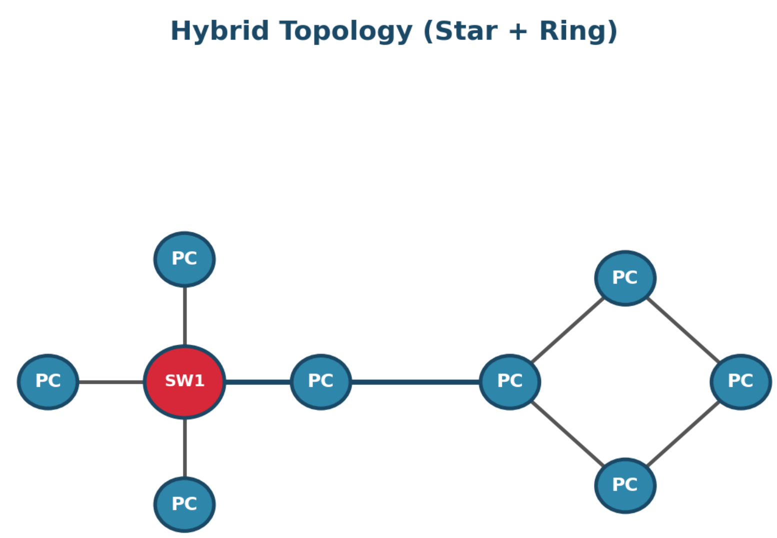

Figure 7: A combination of two or more topologies (here, star and ring)

A **hybrid topology** combines two or more different topologies to leverage their individual strengths and offset their weaknesses – for example, connecting a star network to a ring network, or several star networks joined through a common backbone.

- **Advantages:** Highly flexible and scalable; can be tailored to the specific needs of an organization; more fault-tolerant than a single pure topology.
- **Disadvantages:** More complex to design, manage, and troubleshoot; can be more expensive due to the mixture of equipment required.
- **Use case:** Most real-world large enterprise and campus networks are hybrids in practice.

4.7 Topology Comparison at a Glance

Topology	Cost	Fault Tolerance	Scalability	Best For
Bus	Low	Poor	Poor	Small, temporary, legacy setups
Star	Medium	Good (except hub)	Good	Homes, offices, most modern LANs
Ring	Medium	Fair	Fair	Predictable, orderly traffic (legacy/telecom)
Mesh	High	Excellent	Poor (cost grows fast)	Mission-critical, backbone networks
Tree	Medium-High	Good	Excellent	Large, departmentalized organizations
Hybrid	Varies	Very Good	Excellent	Complex real-world enterprise networks

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Quick Recap

- **Networking** connects devices to share resources and communicate, using nodes, links, and protocols.
- Networks move data as **packets**, addressed and routed by IP/MAC addresses, switches, and routers.
- Networks are classified by **range**: PAN < LAN < CAN < MAN < WAN.
- **LAN** = small, fast, single-owner. **WAN** = large, slower, multi-owner, links LANs together.
- Networks are also classified by **topology**: Bus, Star, Ring, Mesh, Tree, and Hybrid – each with trade-offs in cost, reliability, and scalability.